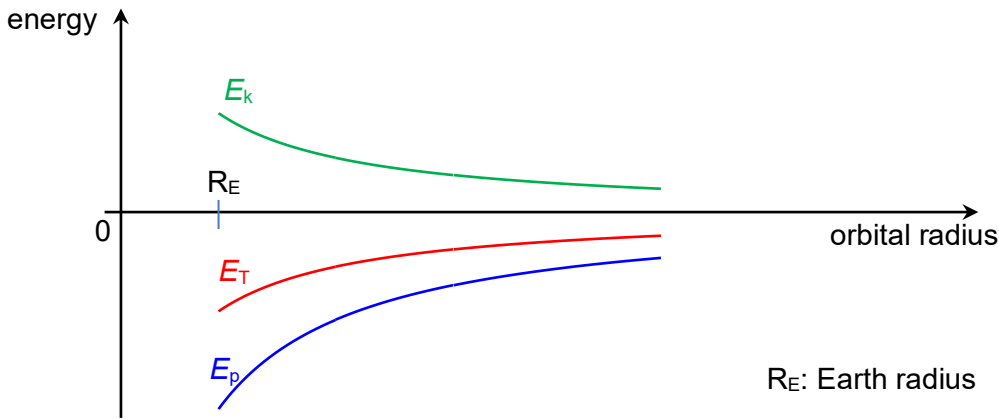
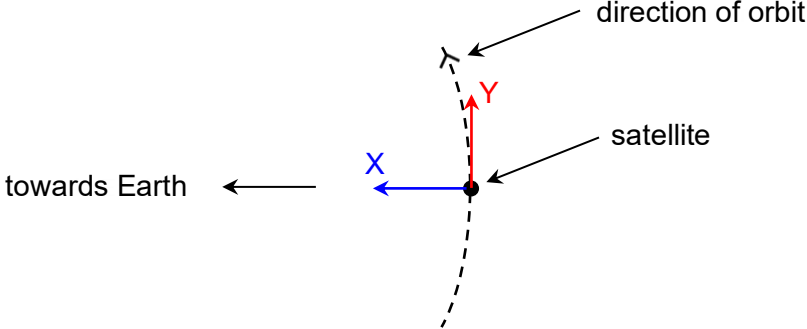


8	(a)	The force of attraction between two point masses is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. 0 mark for nonsensical definition. 1 mark for mostly correct definition. 2 marks for faultless definition.	
	(b)	(i) Gravitational pull provides the required centripetal force. $\frac{GMm}{r^2} = m \frac{v^2}{r}$ $\frac{1}{2} \frac{GM}{r} = \frac{1}{2} mv^2$ $-\frac{1}{2} \left(-\frac{GM}{r} \right) = \frac{1}{2} mv^2$ $-\frac{1}{2} E_p = E_k$ $E_p = -2E_k$ 1 mark for correct GPE or KE. 2 marks for full derivation.	
		(ii)  1 mark if a least 1 graph is correct. 2 marks for all correct.	
	(c)	(i) $\frac{GMm}{r^2} = m \frac{v^2}{r}$ $\frac{GM}{r} = v^2$ $\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.7 \times 10^6} = v^2$ $v = 7716 \text{ m s}^{-1}$ $= 27800 \text{ km h}^{-1}$ M1 at correct substitutions. A1 at correct numerical answer.	
		(ii) At this altitude, the centripetal force provided by the Earth is larger than the required centripetal force (for circular motion at geostationary speed).	B1

	(d)	(i)	GPE is converted into KE and heat. 1 mark if KE or heat is omitted.	
		(ii)	$\frac{\Delta TE}{TE} = \frac{\Delta r}{r}$ $= 0.2\%$	B1
		(iii)	$\Delta TE = -\frac{0.2}{100} \times \left(\frac{1}{2} \frac{GMm}{r} \right)$ $= -\frac{0.2}{100} \times \frac{1}{2} \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(1.80 \times 10^3)}{6.70 \times 10^6}$ $= -\frac{0.2}{100} \times 5.3579 \times 10^{10}$ $= 1.072 \times 10^8 \text{ J}$ Rate of loss of energy = $\frac{1.072 \times 10^8}{7 \times 24 \times 3600} = 177 \text{ J s}^{-1}$ $(P = Fv)$ $177 = f(7716)$ $f = 0.0230 \text{ N}$	M1 M1 M1
	(e)	(i)		B2
		(ii)	$F = uz$ $0.0230 = 2.00 \times 10^3 z$ $z = 1.148 \times 10^{-5} \text{ kg s}^{-1}$ Fuel needed = $24 \times 3600 \times 1.148 \times 10^{-5} = 0.992 \text{ kg}$	M1 A1